



Department of Computer Science and Engineering
Final Examination Summer 2023
CSE 321: Operating Systems

Duration: 1 Hour 50 Minutes

Total Marks: 40

Answer the following questions.
Figures in the right margin indicate marks.

1. a) Imagine a computer lab with multiple computers, and equipped with printers. Students use these computers for various tasks and may want to print documents. However, there are only a limited number of printers available. How do you solve this issue using semaphore? Your answer should have the steps associated in solving the given scenario. [3]

CO4

b) How do semaphores and mutexes differ in their implementation. [1.5]

c) For Peterson's problem below conditions will be applied.

- There are two processes: P1 and P2. P2 gets to execute first.
- Each Statement takes 4ms to execute.
- Context Switch will occur after 12ms.
- Critical section contains 4 statements.
- Remainder section contains 3 statements.
- For P1: i=0 and j=1
- For P2: i=1 and j=0
- turn=0
- flag[0] = FALSE, flag[1] = TRUE

The structure of process P_i in Peterson's solution:

```
do{
    flag[i] = true;
    turn = j;
    while(flag[j] == true && turn == 1){
        //busy wait
    }
    //critical section
    flag[i] = false;
    //remainder section
}while(true);
```

Complete the table below for processes P₁ and P₂ using **Peterson's solution**. [3.5]

Process 1: i = 0, j = 1	Process 2: i = 1, j = 0



2. a) Consider the following snapshot of a system:

CO4

		Allocation			Max			Available		
		A	B	C	A	B	C	A	B	C
P0		5	2	2	9	9	8	7	10	5
P1		3	2	0	9	9	10			
P2		5	3	3	6	8	5			
P3		3	0	0	6	7	9			
P4		4	2	1	5	7	6			

i. **Calculate** the Need Matrix. Is this system in a **safe state**? If yes, then find the safe sequence using Banker's Safety algorithm otherwise, provide the necessary explanation. [1+3]

ii. What happens if the process **P4** requests at this moment for **(0, 3, 1)**? Whether Banker's algorithm grants the request or not? [5]

b) Suppose, in a workplace, we have a set of resource types, $R = \{R1, R2, R3, R4\}$ and a set of processes, $P = \{P1, P2, P3, P4, P5\}$. **R1, R2, R3, and R4** have **3, 1, 4, and 2** instances respectively.

- P1 is holding 2 instances of R1
- P2 is holding 1 instance of R3
- P3 is holding 1 instance of R4
- P5 requests 2 instances of R3
- P4 is holding 1 instance of R4
- P3 requests 1 instance of R2
- P2 requests 1 instance of R1
- P2 is holding 1 instance of R2
- P1 is requesting 1 instance of R4
- P3 is holding 1 instance of R3
- P4 is holding 1 instance of R3.

Construct a resource allocation graph for the above scenario and **identify the cycle (if any) and decide** whether there is a deadlock or not. [4]

3. a) Given variable size memory (dynamic) partitions of **10 MB, 16 MB, 45 MB, and 49 MB** (in order, top to bottom), **apply** best-fit and worst-fit algorithms to place processes with the space requirement of **5 MB, 15 MB, 10 MB, 5 MB, 10 MB, 20 MB, 25 MB, and 15 MB** (in order). Which algorithm makes the most [4+1]

CO5

effective use of memory?

b) How cache memory can be used to design an efficient paging hardware. [3]
Your answer should have the required diagram with the necessary justification.

c) Compare the following systems in terms of performance: [3]

1. A system with a hit ratio of 72%, associative lookup time of 24ns, and memory access time of 56ns.
2. A system with a hit ratio of 65%, associative lookup time of 6ns, and memory access time of 133ns.

4. **a)** Suppose there are two processes **P1 (32 B)** and **P2 (40 B)** with a page size of **8 B**. The main memory size is **96 B**. The page table for P1 and P2 are given below:

CO5

Page table: P1

Page table: P2

Find the corresponding physical addresses of the following logical addresses: [3]

- a) Address 000001 of P1
- b) Address 001001 of P2
- c) Address 101001 of P2

b) Consider a computer with a main memory that has 5 frames and page reference string of 0-7 page [6, 5, 3, 6, 4, 5, 0, 5, 5, 1, 1, 5, 0, 5, 4, 6, 3, 5, 1, 5]. The page reference string represents the order in which the pages are accessed by a program. **Apply LRU & OPT** algorithm to **simulate** the page replacement that occurs when the main memory can hold at most 5 pages at a time. **Record** the number of **page faults** and compare the result. **Mention** which algorithm performs better in this scenario. [4+1]